

# Rafael A. Dutra

Junior AI Developer | Machine Learning Engineer

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## PROFESSIONAL SUMMARY

Junior AI Developer seeking first full-time opportunity in a high-performance environment. With hands-on production experience in LLMs, multi-agent systems, RAG and Machine Learning, I worked at the Porto Digital/Globo Technology Residency architecting ETL pipelines and supervised models (F1 >0.90, ROC-AUC >0.92) for automatic anomaly detection in live broadcasts — reducing operational costs and accelerating business decisions. Experience in Data Analysis, Power BI and the full ML lifecycle: data collection, modeling, containerization and monitoring. English C2 (EFSET).

## TECHNICAL SKILLS

- **IA & Machine Learning:** LLMs (GPT-4o), RAG, Autonomous Agents, NLP, Prompt Engineering, Computer Vision, Scikit-learn, TensorFlow, PyTorch, XGBoost, SHAP
- **Data & Analytics:** ETL, Data Pipelines, Data Cleaning, Business Insights, Power BI, SQL, Pandas, NumPy, Streamlit, Jupyter, Excel
- **Languages & Frameworks:** Python, JavaScript, FastAPI, React, Biopython
- **Infra & Data:** PostgreSQL, MongoDB, Redis, Docker, Docker Compose, Git/GitHub, Linux, REST APIs

## PROFESSIONAL EXPERIENCE

**AI Resident Engineer — Technology Residency Porto Digital / Globo — Recife, Brazil** Feb 2024 — Present

- Architected Horus AI, an automatic fault detection system for live broadcasts using computer vision and real-time ML, eliminating undetected broadcast errors.
- Built ETL pipelines end-to-end (collection, data cleaning, transformation and persistence) that reduced monitoring operational costs by ~60%, freeing the team for higher-value tasks.
- Orchestrated supervised models with F1-Score >0.90 and ROC-AUC >0.92 in production, generating business insights and automated alerts with direct impact on broadcast quality.

## FEATURED PROJECTS

**MedNode — Clinical Decision Support Multi-Agent System** • Python · FastAPI · PostgreSQL · React · Docker · GPT-4o-mini 2024 — in development

- Architecting a system with 3 orchestrated autonomous agents: ClinicalAgent (symptom/image analysis via GPT-4o-mini Vision), PubMedAgent (RAG with fallback across 4 scientific APIs) and Orchestrator (evidence-based synthesis).
- Implementing API cost-control middleware with per-user quotas and granular logging, enabling a freemium model. Containerized deploy with Docker + Traefik, automatic SSL and JWT authentication.

**Horus AI — Live Stream Anomaly Detection** • Python · FastAPI · TensorFlow · OpenCV · Angular · PostgreSQL · WebSocket 2024 — Present

- 3 parallel ML models: video (Odin v4.5, 97.6%), audio (Heimdall Ultra v1, 90.9%) and lipsync (SyncNet v2, 100%) analyze SRT streams in real time, detecting freeze, fade, blur, silence, echo and desync.
- Hybrid strategy combining OpenCV heuristics + ML with temporal voting; automatic clip generation as evidence, real-time WebSocket alerts, KPI dashboard and PDF report export.

## EDUCATION

**Systems Analysis and Development — FICR (Faculdade Imaculada Conceição do Recife)** Feb 2024 — Jul 2026  
**Computer Science Degree (incomplete) — UFRPE (Federal Rural University of Pernambuco)** 2019 — 2022

## LANGUAGES

**Portuguese:** Native • **English:** C2 Proficient — EFSET Certified • **Spanish:** Basic

## CERTIFICATIONS

English C2 Proficient (EFSET) • ONE Tech Foundation G8 — Data Science + AI (Oracle/Alura) • SQL Advanced (Kaggle) • Python (Kaggle)